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Impact of frontline employees' perceived benefits of artificial intelligence on AI adoption and job engagement: A meta-analysis

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ABSTRACT

This study explores how frontline employees' perceived benefits of artificial intelligence (AI) influence two behavioural outcomes: AI adoption and job engagement. Using the Social Shaping of Technology (SST) framework, which highlights the role of social, organizational, and contextual factors over technical features, the meta-analysis synthesises 71 studies involving 23,051 employees across 37 countries. Perceived AI benefits (e.g., increased efficiency, reduced monotony, decision-making support) positively relate to both adoption and engagement, though risks like job displacement anxiety can weaken these effects. To identify when and for whom these benefits matter most, the study analyses 13 moderators (user-level: gender, education, age, tenure, role, position; organizational-level: industry, control, AI task and knowledge; macrolevel:year, country). Industry type, control over AI, and work level/position were most significant. The findings underscore the need for tailored AI strategies aligned with workforce characteristics and socio-technical contexts, advancing employee-centred AI research on contextual variation and moderating effects.

1. Introduction

The rapid growth of artificial intelligence (AI) is transforming workplaces, especially in the hospitality sector, where delivering exceptional consumer experiences is vital (Ma et al., 2024). Organizations are increasingly adopting AI tools to streamline operations, enhance service personalisation, and support customer engagement (Dequito, 2024; Schwarz, 2025). In hospitality industry, AI improves operational efficiency, decision-making, and automates repetitive tasks such as scheduling, guest messaging, and service customisation (Behnke, 2025; Presbitero and Teng-Calleja, 2023). For frontline service employees, AI reduces routine workload, allowing focus on customer-centric and complex responsibilities (Song et al., 2022c). However, challenges like job engagement and AI adoption persist (Vrontis et al., 2023). Concerns about job automation and insecurity (Braganza et al., 2021; Chuang, 2021; He et al., 2023) remains particularly pronounced in hospitality settings, such as hotels and restaurants, where the rise of AI introduces tension between the need for human interaction and the push for automation (Parvez et al., 2022; Leung et al., 2023).

Successful AI integration requires aligning frontline employees' demographic, technical, cognitive, and emotional competencies with AI system requirements (Zirar et al., 2023). Demographic factors like age, gender, and education influence training needs (Wijayati et al., 2022). Younger employees adapt more easily, while older workers often require extra support (Dutta et al., 2022; Fu et al., 2024). Technical proficiency enables effective AI use (Kang et al., 2023), while emotional readiness fosters engagement (Suseno et al., 2023; Zhu et al., 2023). Clear communication, support systems, and active participation are critical (Suseno et al., 2023).

Although AI benefits like engagement and performance gains are noted, risks such as job insecurity and self-esteem issues may trigger resistance (He et al., 2023; Tojib et al., 2023). Therefore, from the viewpoint of frontline employees, a holistic understanding of AI's perceived benefits and risks is essential to encourage job engagement and facilitate AI adoption.

This meta-analysis addresses a critical gap by investigating the relationship between perceived AI benefits and frontline employees'

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behavioural responses (i.e., adoption and engagement) within the hospitality industry (Chowdhury et al., 2022). While adoption represents initial use, engagement reflects sustained integration (Alam et al., 2024). Despite AI's increasing presence in hospitality, current research is fragmented and lacks synthesis (Park et al., 2024; Parvez et al., 2022). Studies report mixed findings: some highlight AI's potential to reduce monotony and enhance employee satisfaction (Nguyen and Malik, 2022a), while others caution that displacement and reskilling concerns may drive turnover (Suseno et al., 2023; Zhu et al., 2023). Moreover, most hospitality research has focused on customer experiences and service quality, while employee-centric outcomes regarding persistent challenges of job engagement and adoption of AI remain underexplored (Qiu et al., 2022). Existing meta-analyses have largely excluded frontline service contexts or failed to account for unique industry characteristics (Ashaduzzaman et al., 2023; Gao et al., 2016; Ma et al., 2024).

Accordingly, this meta-analysis aims to answer: *"How do perceived benefits of AI drive job engagement and AI adoption among frontline service employees, and under what conditions are these effects strengthen or weaken?"*

To do so, this study examines moderating factors such as employees' AI control, task knowledge, gender, education, age, job roles, as well as the industry context (e.g., service vs. non-service sectors), which shape how employees perceive AI benefits and their behavioural responses (Chandra and Rahman, 2024). Understanding these moderators offers insights into tailoring strategies that foster trust, alignment, and positive behavioural responses during AI integration (Shamim et al., 2023).

The analysis provides theoretical and practical insights into fostering positive employee behaviours during AI integration (Ashaduzzaman et al., 2023; Wang et al., 2024). Adoption ensures tool integration, while engagement aligns employees with organizational goals (Suseno et al., 2023). While AI supports functions like scheduling, guest profiling, and service personalisation (Rodgers et al., 2023; Delcker et al., 2024), its success hinges on employee trust, participation, and ongoing engagement (Kang et al., 2024). The next sections provide literature review and hypotheses, methodology, results, discussion and conclusion respectively.

2. Literature review and hypothesis development

2.1. Definition of perceived benefits of AI

The integration of AI in service environments reshapes frontline employees' psychological responses, influencing AI adoption and job engagement (Braganza et al., 2021; Li et al., 2024a). Although AI offers clear advantages in hospitality, barriers like job automation anxiety hinder adoption (Mirbabaie et al., 2022). Perceived benefits, including efficiency, task support, and transparency, affect how employees view AI's alignment with job goals, fostering adoption and engagement (Williams and Edge, 1996; Russell and Williams, 2002; Pillai et al., 2024; Zhu and Kanjanamekanant, 2023). For example, recognising AI's efficiency and service compatibility enhances positive attitudes (Pillai et al., 2024). Employees seeing AI as supportive, not controlling, feel more empowered and engaged (Wang et al., 2023b). Thus, perceived benefits drive both AI adoption and sustained job engagement (Park et al., 2024).

2.2. Theoretical framework

The Social Shaping of Technology (SST) framework provides a comprehensive lens to understand how technology adoption, including AI, unfolds within organizations (Mackay and Gillespie, 1992). Unlike deterministic models that portray technology as an external force acting upon users, SST posits that adoption emerges from dynamic interactions among social processes, user interactions, and contextual factors (Howcroft and Taylor, 2023). SST encompasses the social, organizational, and contextual dimensions that influence how technology is interpreted, adopted, and integrated into workplace practices (Kim and Kim, 2024).

Compared to Technology Acceptance Model (TAM) or the Unified Theory of Acceptance and Use of Technology (UTAUT), which emphasise individual-level cognitive constructs such as perceived usefulness and ease of use, SST framework offers a broader and more nuanced perspective (Madan and Ashok, 2024; Williams and Edge, 1996). It accounts for organizational norms, power dynamics, emotional responses, and broader societal expectations (Mackay and Gillespie, 1992; Russell and Williams, 2002). Rather than treating adoption as a deterministic outcome of technological features, it highlights the complex and context-dependent nature of how technologies are adopted and used (Madan and Ashok, 2024). This makes SST particularly valuable for understanding AI integration in frontline service environments, where it can significantly reshape job roles, autonomy, and workplace dynamics (Williams and Edge, 1996).

SST frames perceived AI benefits as socially constructed beliefs influenced by individual, organizational, and societal factors. These factors shape AI adoption and engagement by supporting control over AI, job roles, industry type, and emotional readiness, which are typically overlooked in more rigid models (Park et al., 2024; Russell and Williams, 2002; Wong et al., 2023).

Guided by SST, this study examines how frontline employees' behavioural responses (i.e., AI adoption and job engagement) are influenced not only by the technical features of AI but also by the alignment between AI systems and employees' social and organizational contexts (Wang et al., 2023b; Williams and Edge, 1996). This approach supports the inclusion of hypotheses that consider the moderating effects of demographic (e.g., age and gender), organizational (e.g., job role and work level), and contextual (e.g., industry type and country) variables. These considerations reflect SST's view that technology outcomes depend on local interactions among individuals, norms, and institutional structures (Mackay and Gillespie, 1992; Russell and Williams, 2002).

Appendix A lists the variables used and their aliases. This framework synthesises the antecedents, moderators, and outcomes of perceived AI benefits using SST to analyse their dynamics. (see Fig. 1).

2.3. Relationship between perceived benefits of AI and AI adoption

AI adoption refers to the extent to which frontline employees incorporate AI into workflows (Presbitero and Teng-Calleja, 2023). Despite expanding AI use, adoption remains low due to limited formal support systems, as only 21 % of organizations with AI integration have such policies (McKinsey and Company, 2023). Employees preferring routine often struggle with disruptions AI introduces (Frick et al., 2021; Suseno et al., 2023). However, recognising AI's efficiency, error reduction, and repetitive task minimisation improves adoption willingness (Verma and Singh, 2023). Positive perceptions of AI benefits significantly enhance employees' readiness to use it (Pillai et al., 2024; Sattu et al., 2024). By automating mundane tasks, AI allows employees to focus on complex activities, fostering favourable attitudes (Verma and Singh, 2023).

H1. : *Perceived benefits of AI positively correlate with AI adoption.*

2.4. Relationship between perceived benefits of AI and job engagement

Job engagement is a work-related psychological state characterized by vigour, enthusiasm, and immersion (Schaufeli et al., 2006; Kong et al., 2021). It is considered vital for organizational performance and productivity (Prentice et al., 2023). High engagement often stems from organizational support and recognition of employees' contributions (Wu and Zhang, 2024). Perceiving AI as enhancing performance strengthens employees' sense of value, competence, and empowerment (Kong et al., 2021; Zirar et al., 2023). In hospitality, monotonous and

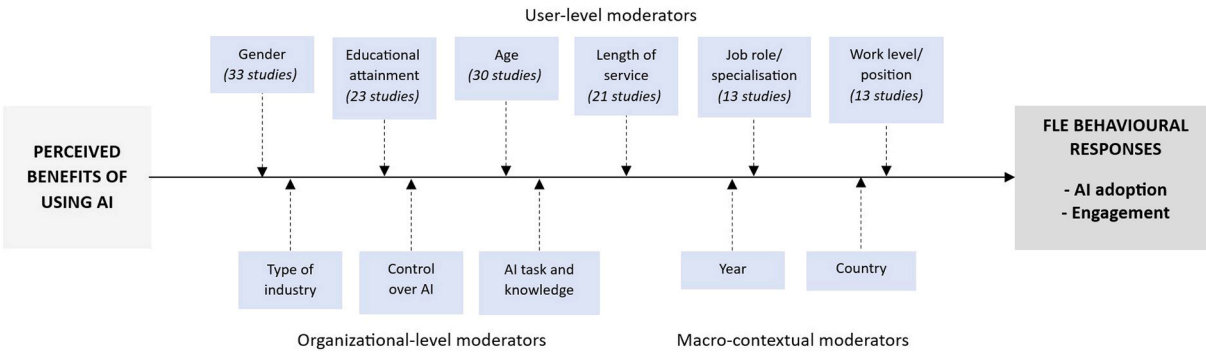

Fig. 1. Meta-analysis framework.

time-consuming tasks hinder efficiency and satisfaction (Qiu et al., 2022). AI reduces repetitive, error-prone activities, improving operational efficiency and job satisfaction (Zhu and Kanjanamekanant, 2023). This fosters a sense of accomplishment and purpose, key elements of engagement. Employees viewing AI as enhancing well-being and work experience exhibit higher engagement (Wang et al., 2023b).

H2. : *Perceived benefits of AI positively correlate with job engagement.*

2.5. Moderators

Research highlights variations in findings related to moderators, emphasising the need for moderation analysis to understand conflicting results (Shankar et al., 2024). Variables such as year, country, industry type, AI control, AI task and knowledge, gender, educational attainment, age, tenure, job role, and work level are hypothesised to moderate the relationship between perceived AI benefits and frontline employees' behavioural responses (de Oliveira Santini et al., 2023). In line with SST framework, moderators are categorised into three domains: (1) user-level, (2) organizational, and (3) macro-contextual moderators. Table 1 presents this classification, and Appendix B details the coded characteristics of the reviewed literature.

Table 1
Moderators.

Research Design Constructs	Operationalisation	Coding (number of studies)
User-level		
Gender *Male* *Female*	Whether the percentage of male participants was greater in the sample set, or the percentage of female participants was greater in the sample set	
Educational attainment	Whether the respondents' educational attainment are foundation/vocational courses, bachelors, or postgraduates	0 =foundation/ vocational; 1 =bachelors; 2 =postgrad
Age		average age in the sample
Length of service		average length of service in the sample (no. of years)
Job role	Whether the data is collected from frontline employees, back-office workers, or both	0 =frontline; 1 =back office; 2 =both;
Work level/ position	Whether the respondents within the study are employees with non-managerial or with managerial position in their respective organizations	0 =non-managerial, 1 =managerial
Organizational		
Type of industry	Whether the study is focused on the service delivery environment, non-service, or both	0 =service sector; 1 =non-service sector; 2 =both; 3 =not stated
Control over AI *More control* *Less or no control* *both*	Whether the respondents within the study are employees with high control, agency, and troubleshooting capabilities on AI, no/less control, agency, and troubleshooting capabilities, or mixed	0 =more control; 1 =less or no control; 2 =both
AI task and knowledge *Task-based* *Knowledge based*	Whether the data is collected from task-based respondents/ organization, or knowledge-based respondents/organization	0 =task based; 1 =knowledge based; 2 =both
Macro-contextual		
Year		Year the papers were published
Country	Whether the data is collected from a developed or developing country	0 =developed; 1 =developing; 2 =both

2.5.1. User-level moderators

User-level moderators are individual characteristics of employees that may influence how they adopt, interact with, and respond to AI (Russell and Williams, 2002; Williams and Edge, 1996). In the SST framework, these factors reflect the role of human agency in shaping technology outcomes:

2.5.1.1. Gender. Research shows significant variations in AI perception by gender representation (Dutta et al., 2022). Gender influences behavioural responses to AI, with men often express higher self-efficacy toward performance-driven technologies, while women emphasise fairness and bias mitigation (Liang et al., 2022; Zhu et al., 2023). Yet findings remain inconclusive in hospitality, with some studies showing gender-neutral adoption rates (Braganza et al., 2021; Koo et al., 2021; Mirbabaie et al., 2022). This inconsistency warrants testing gender as a moderator.

H3a:. *Gender moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in men.*

H3b:. *Gender moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in men.*

2.5.1.2. Educational attainment. It refers to the highest level of education completed by an individual (Kong et al., 2021). It shapes attitudes, behaviours, and capacity for knowledge acquisition (Lim and Ok, 2021). Education levels shape frontline employees' trust, job satisfaction, and engagement with organizational learning, thereby influencing their perceptions of and readiness for AI adoption (Lin et al., 2024; Song et al., 2022c). Bachelor's degree holders are more likely to perceive AI as a productivity enhancer and exhibit stronger engagement when supported with structured learning environments (Abdelhakim et al., 2023). However, over-qualification may reduce AI enthusiasm in routine jobs, suggesting a nonlinear relationship (Kong et al., 2021).

H4a:. *Educational attainment moderates the relationship between*

perceived benefits of AI and AI adoption, with a stronger effect in bachelor's degree holders.

H4b:. *Educational attainment moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in bachelor's degree holders.*

2.5.1.3. Age. Chronological age influences work outcomes directly and indirectly (Fu et al., 2024). Frontline employees' attitudes, ease of use, and experiences with AI vary across age groups, affecting job satisfaction (Li et al., 2024b), productivity (Tong et al., 2021), and well-being (Loureiro et al., 2023). Younger employees generally report greater ease with AI and faster adoption, driven by higher digital literacy and familiarity with technology (Fu et al., 2024). However, age is also associated with work experience, enabling older employees to apply procedural knowledge when engaging with AI (Dutta et al., 2022). This suggests that while younger employees may adopt AI more quickly, older employees can derive value when AI is positioned as a complement to their expertise rather than a replacement (Fu et al., 2024). Consequently, the moderating effect of age may depend on how AI applications are framed and supported in the workplace.

H5a. : *Age moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in younger employees.*

H5b. : *Age moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in younger employees.*

2.5.1.4. Length of service. Tenure correlates with commitment and role identity, which can either amplify or inhibit AI engagement depending on perceived role threat (Wijayati et al., 2022). Research suggests that employees with longer tenure often possess greater accumulated skills, experience, and knowledge, which can influence their adaptability to AI (Li et al., 2024b; Zhu et al., 2023). While they may initially resist automation due to status quo bias, adequate training and support can enable them to become key advocates for AI integration (Dutta et al., 2022; Le et al., 2024). The extent to which tenured employees engage with AI may thus depend on the perceived alignment between AI and their accumulated skills, as well as the availability of targeted training and participative implementation efforts.

H6a. : *Length of service moderates the relationship between perceived benefits of AI and AI adoption, with a* stronger *effect in tenured workers*.

H6b. : *Length of service moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in tenured workers*.

2.5.1.5. Job role/specialisation. An employee's position defines their responsibilities and shapes how they perceive AI adoption, particularly in hospitality where frontline roles face direct service impacts (Khaliq et al., 2022; Preston, 2022). Frontline employees, due to customer-facing responsibilities, experience distinct psychological responses than back-end roles, which often face greater job insecurity but benefit from AI efficiency and performance improvements (Malik et al., 2021; Lingmont and Alexiou, 2020). Including back-end roles in this meta-analysis offers comparative insights into AI's effects on frontline employees (Nguyen and Malik, 2022a).

H7a. Job role moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in frontline roles.

H7b. Job role moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in frontline roles.

2.5.1.6. Work level/position. Higher positions establish trust-based, long-term organizational relationships, benefiting from career security, advanced AI-related skills, and appreciation of AI integration benefits (Xu et al., 2023b). Research shows higher organizational hierarchy positions correlate with stronger organizational attachment (Wang et al., 2022). Full-time employees exhibit greater commitment due to employer support and advancement opportunities, while part-time workers demonstrate less commitment due to limited inclusion in organizational systems (Huang et al., 2024). Frontline employees in higher positions within hospitality experience greater job variety and autonomy, fostering stronger commitment compared to lower-level roles (Rabiul et al., 2023). Therefore, hierarchical position can condition the strength of AI benefit perceptions on behaviour.

H8a. Work level moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in managerial roles.

H8b. Work level moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in managerial roles.

2.5.2. Organizational-level moderators

Organizational-level moderators refer to workplace factors that shape how technologies are implemented and experienced (Williams and Edge, 1996). In SST, these reflect how organizational structures influence technology use (Mackay and Gillespie, 1992; Russell and Williams, 2002).

2.5.2.1. Type of Industry. Service industries like healthcare and hospitality benefit from AI's role in improving customer interactions and decision accuracy (Parvez et al., 2022; Wang et al., 2022). Non-service industries prioritise efficiency, with distinct outcomes and challenges in AI adoption such as displacement (Nguyen and Malik, 2022a). Service sectors leverage AI to enhance customer-centric goals, increasing engagement and perceived benefits (Lin et al., 2022).

H9a. Type of industry moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in the service sector than in the non-service sector.

H9b. Type of industry moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in the service sector than in the non-service sector.

2.5.2.2. Control over AI. It refers to frontline employees' ability to manage and direct AI in alignment with objectives or regulations (Liang et al., 2022; Chowdhury et al., 2022). Autonomy enhances engagement and adoption, while lack of control, especially in rigid roles, can lead to resistance and diminished role identity (Mirbabaie et al., 2022; Wang et al., 2022). Employees with greater input into AI behaviour and access to training and feedback systems report higher perceived control and smoother integration (Mirbabaie et al., 2022; Wang et al., 2023b; Wu and Zhang, 2024). Thus, the ability to influence AI outcomes may significantly moderate perceived benefit effects.

H10a. : *Control* moderates *the relationship between AI benefits and adoption, with stronger effects in employees with more control.*

H10b. : *Control moderates the relationship between AI benefits and job engagement, with stronger effects in employees with more control.*

2.5.2.3. AI task and knowledge. AI's knowledge capabilities, such as data analysis and decision-making, enhance service delivery efficiency and innovation, fostering well-being, trust, and role confidence during AI integration (Cao et al., 2021; Koo et al., 2021; Chowdhury et al., 2022; Prentice et al., 2023). Systems with robust knowledge capabilities empower employees to recognise AI's benefits, alleviating job displacement concerns (Yu and Li, 2022). Task-oriented AI, focused on executing specific functions, streamlines routine activities and enhances productivity, allowing employees to prioritise complex tasks requiring empathy or intuition (Song et al., 2022b; Shamim et al., 2023). Distinguishing task-based and knowledge-based AI enables effective task allocation (Chowdhury et al., 2022; Verma and Singh, 2023; Wang et al.,

2024).

H11a. AI task and knowledge moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in task-based capabilities.

H11b. AI task and knowledge moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in task-based capabilities.

2.5.3. Macro-contextual moderators

Macro-contextual moderators include broader environmental factors that represent how societal norms, policies, and historical context shape the perception and adoption of AI (Russell and Williams, 2002; Williams and Edge, 1996), consistent with SST principles:

2.5.3.1. Year. Year of publication reflects AI integration's rapid evolution, particularly during the COVID-19 pandemic (Leung et al., 2023; Parvez et al., 2022; Shankar et al., 2024). Research from 2018 to 2020 showed modest activity, with a significant increase in 2021 due to remote work and social distancing measures (Hu et al., 2021; Wijayati et al., 2022). Greater exposure to AI enhances employees' comfort, engagement, and receptivity to AI benefits (Yang et al., 2024). Outcomes evolve with AI's expanding applications, highlighting the importance of timely research (Shamim et al., 2023). Thus, year moderates the salience and framing of AI benefits.

H12a. Year moderates the relationship between AI benefits and adoption, with stronger effects in recent years.

H12b. Year moderates the relationship between AI benefits and job engagement, with stronger effects in recent years.

2.5.3.2. Country. Trust and acceptance of AI vary between developed and developing countries due to differences in institutional support, innovation capacity, and inter-firm collaboration (Abdelhakim et al., 2023; Ashaduzzaman et al., 2023; Shamim et al., 2023). Developed nations often leverage internal resources and policy infrastructure for digital transformation, while developing economies face barriers such as limited funding, training gaps, and language-related access to knowledge (Ashaduzzaman et al., 2023). Existing AI studies focus on Western and Asia-Pacific regions (Dang and Liu, 2022; Nguyen and Malik, 2022b), with underrepresentation of Arab and Southeast Asian contexts, where adoption remains challenging (Shamim et al., 2023; Shankar et al., 2024). These disparities suggest that country context moderates the impact of perceived AI benefits on employee behaviour.

H13a. Country moderates the relationship between AI benefits and adoption, with stronger effects in developed nations.

H13b. Country moderates the relationship between AI benefits and job engagement, with stronger effects in developed nations.

3. Methodology

Meta-analysis estimates population effects' mean and variance from multiple studies and is widely used in research and review articles (Field and Gillett, 2010). Variability in effect sizes, common in studies (Field, 2003), can be examined through moderating variables (Field, 2005; Li et al., 2024b). The step-by-step process of this meta-analysis is outlined below.

3.1. Literature search

A literature search was conducted using six approaches. First, a keyword search of scholarly databases (Web of Science, Science Direct, Emerald, ProQuest, Sage, Taylor and Francis, and Springer) using keywords like "artificial intelligence," "AI benefits," "perceived benefits of AI," "frontline," and "employees" was performed were chosen based on their relevance in prior meta-analyses and frequent use in empirical studies focused on AI and frontline work. (Appendix C) (Castaneda et al., 2025; Harari et al., 2020; Le et al., 2024). Second, relevant theses and dissertations were sourced from ProQuest. Third, unpublished works were identified via Social Science Citation Index (SSCI) and Social Science Research Network (SSRN) (Faramarzi et al., 2024). Fourth, marketing conference proceedings (e.g., ANZMAC) were reviewed. Fifth, reference lists from AI-related reviews (e.g., (Madan and Ashok, 2024)) were examined. Finally, Google Scholar was used for verification (Adil et al., 2022). This process identified 1024 papers.

3.2. Inclusion and exclusion criteria

Relevant articles were selected using specific criteria (Maseeh et al., 2022). Eligible articles must (i) be in English, (ii) qualify as scholarly articles, (iii) focus on AI's perceived benefits on frontline employees' behavioural responses, (iv) employ quantitative methods, and (v) report correlation coefficients or convertible statistics like F ratios. From 1024 papers, 537 duplicates were removed, leaving 487 unique articles. Further exclusions included: 16 non-English papers, 198 non-scholarly sources, 113 studies unrelated to the central theme, 57 non-quantitative articles, and 32 without usable statistical data, resulting in 71 studies eligible for meta-analysis (see Fig. 2).

3.3. Coding

The final dataset consisted of 71 studies covering 23,051 frontline employees from 37 countries. Coding was performed using Microsoft Excel. Variables with similar conceptual meanings (e.g., job crafting (Song et al., 2022a) and work engagement (Ding, 2021)) were grouped accordingly based on prior research (Schaufeli et al., 2006). Variables were coded based on conceptual meanings from literature (Maseeh et al., 2021). Two independent coders conducted the process, and discrepancies were resolved through discussion. Inter-rater reliability was measured using Krippendorff's alpha. The resulting value of 0.894 exceeds the commonly accepted threshold of 0.80, indicating high consistency between the independent coders during the classification of variables (Krippendorff, 2013).

3.4. Analytical steps

3.4.1. Publication bias examination

To ensure robustness of findings and validity of conclusions drawn from meta-analytic research (Egger et al., 1997; Li et al., 2024b), publication bias, also known as the file-drawer problem, was examined using three statistical techniques: the funnel plot method (Blut et al., 2023), Egger's test (Ceylan et al., 2024; Egger et al., 1997), and fail-safe N method (Rosenthal, 1979; Faramarzi et al., 2024). Funnel plots visualised the distribution of zero-order correlations on the X-axis against the precision index on the Y-axis. Asymmetry in these plots may suggest potential publication bias (Shankar et al., 2024) (Fig. 3).

The fail-safe N method quantified the number of additional studies required to nullify meta-analysis results (Maseeh et al., 2021), with a bivariate relationship considered free of bias if the fail-safe N value exceeded $5k + 10$ (Chen et al., 2024). Egger's test analysed the linear relationship between effect size estimates and their standard errors (Maseeh et al., 2022; Mehta et al., 2022), where a significant p-value indicated bias and an insignificant p-value suggested its absence (Blut et al., 2023).

3.4.2. Bivariate analysis

Zero-order correlations served as the primary effect size metric (Shankar et al., 2024). When bivariate correlations were unavailable in empirical studies, alternative effect sizes were used to calculate zero-order correlation coefficients (Pu et al., 2024). Mean correlations

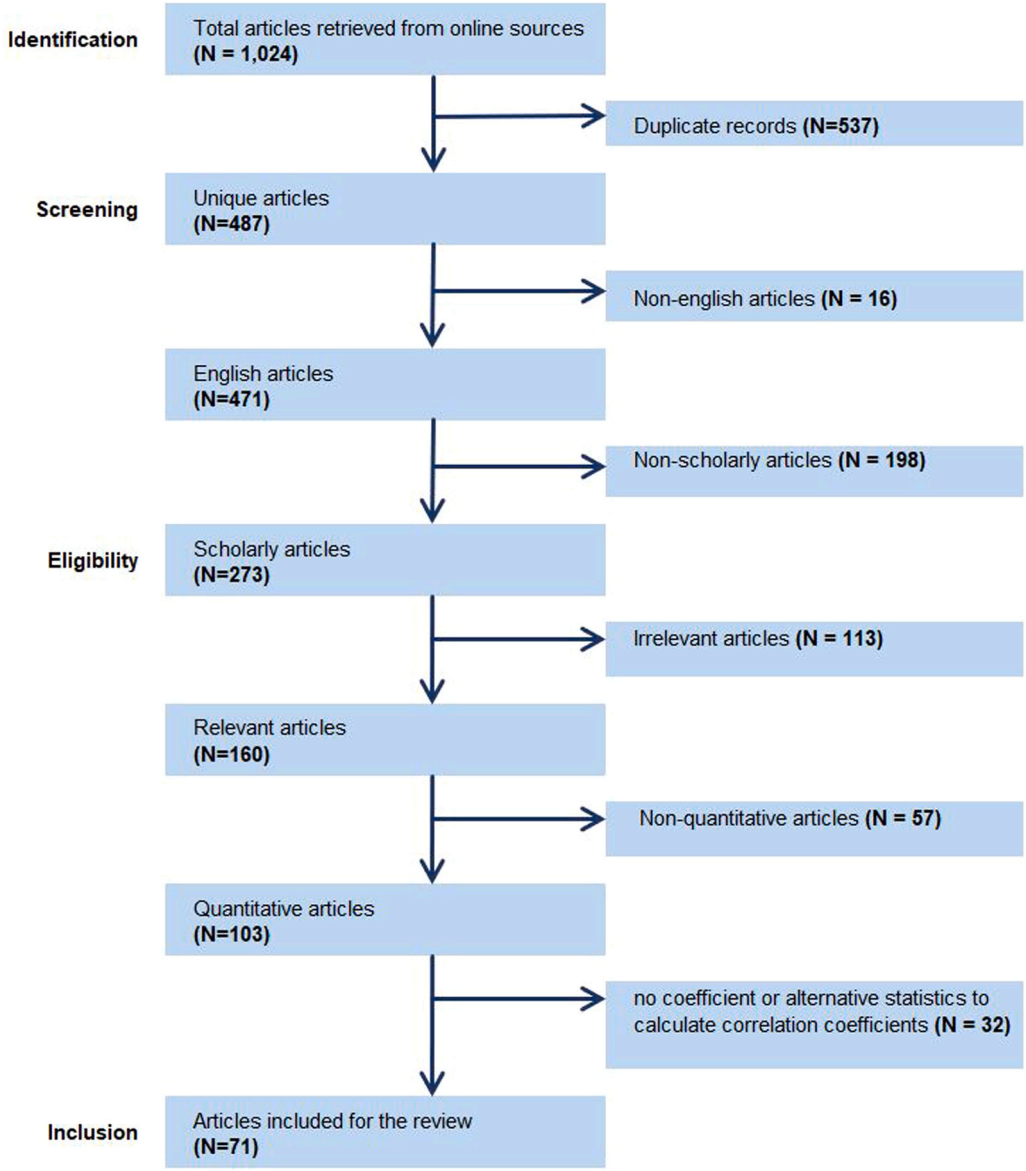

Fig. 2. article selection process.

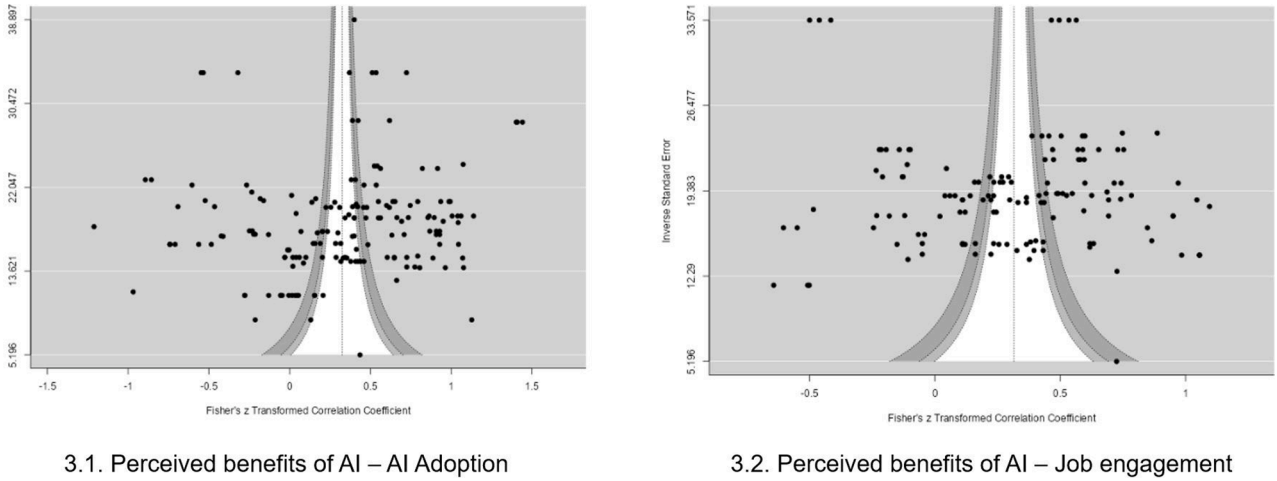

Fig. 3. Funnel plots.

(r +) and reliability-corrected correlations (ρ) were calculated to assess bivariate relationships (Zhu et al., 2023). A random-effects approach adjusted zero-order correlations by dividing them by the product of the square roots of Cronbach's alpha (α) (Good et al., 2022). This process involved dividing the zero-order correlations by the product of the square roots of Cronbach's alpha (α). If α was unavailable, composite reliability was used (Yee et al., 2022).

Adjusted correlation coefficients were converted into Fisher's Z (Mehta et al., 2022; Zhu et al., 2023), then back to zero-order correlations by averaging them, weighted by inverse variance estimates. The significance of adjusted correlation coefficients (ρ) was determined using 95 % Confidence Intervals (CI) (Teng et al., 2024). A corrected correlation coefficient (ρ) was significant if its CIs excluded zero (Michel et al., 2023).

3.4.3. Assessment of heterogeneity

To determine whether the effects were consistent across studies, heterogeneity was assessed using Q and I^2 statistics (Maseeh et al., 2021). Significant Q values indicate true heterogeneity, while I^2 quantifies the percentage of total variance not attributable to sampling error (Mehta et al., 2022; Ma et al., 2024).

3.4.4. Moderation analysis

Meta-regression analysis examined the moderating effects of user-level, organizational, and macro-contextual variables on the relationships between perceived AI benefits and employee behavioural outcomes (Zhu et al., 2023) (Table 2).

Several moderators (i.e., educational attainment, age, length of service, job role, work level/position, control over AI, AI task and knowledge, year and country) showed varying significance. For instance, educational attainment negatively moderated adoption, while age positively moderated adoption, and length of service enhanced engagement (Frick et al., 2021; Suseno et al., 2023). Notably, control over AI negatively moderated both outcomes, contrary to prior assumptions (Loureiro et al., 2023; Wang et al., 2022). Other moderators (i.e., gender, age on engagement, and length of service on adoption), were non-significant, suggesting these demographic factors do not influence how employees translate perceived AI benefits into behavioural outcomes.

A comprehensive interpretation of these findings and their practical implications for hospitality managers is provided in the managerial implications. Understanding these patterns is essential for business and HR leaders aiming to tailor AI implementation strategies to maximise employee adoption and engagement (Kambur and Akar, 2022).

3.4.5. Multicollinearity diagnostics

To confirm multicollinearity did not impact meta-analysis results, researchers conducted rigorous checks during multilevel meta-regression (e.g., moderation analysis). Inter-correlations among constructs were reviewed, excluding coefficients exceeding 0.80. Variance inflation factor (VIF) estimates, all below 5, indicating no multicollinearity issues (Luffarelli et al., 2023). Sensitivity analyses were conducted to test robustness including leave-one-out analysis, where one construct at a time with coefficients over 0.50 was excluded (Steel et al., 2021). Consistent results further verified multicollinearity did not disproportionately influence the findings (Shankar et al., 2024). While study quality was not scored explicitly, funnel plot symmetry and Egger's test provided indirect evidence that study variance did not distort overall outcomes.

4. Results

4.1. Descriptive statistics, file drawer, and heterogeneity diagnostics

Table 3 presents descriptive statistics, indicating significant positive associations of perceived benefits with AI adoption (ρ=0.324***) and job engagement (ρ=0.315***), supporting H1 and H2. Funnel plots assessed publication bias for both bivariate relationships (Fig. 3).

Table 2
Moderation analysis.

Bivariate relationship		No. of studies	Perceived Benefits of AI – AI adoption					Perceived Benefits of AI – Job engagement				
			β	SE	Z-value	$CI_{95\,\%}$	VIF	β	SE	Z-value	$CI_{95\,\%}$	VIF
Intercept			184.606***	14.244	12.959	[156.687; 212.525]		−53.803	68.412	−0.785	[−187.889; 80.282]	
User-level												
Gender	H3	57	0.289	0.193	1.497	[−0.089; 0.668]		0.140	0.220	0.638	[−0.291; 0.573]	
Educational attainment	H4	47	−0.046*	0.018	−2.553	[−0.082; −0.010]	1.020	−0.034	0.105	−0.328	[−0.239; 0.171]	1.011
Age	H5	55	0.012***	0.001	9.296	[0.009; 0.013]	1.000	−0.003	0.004	−0.815	[−0.012; 0.005]	1.000
Length of service	H6	28	0.020	0.018	1.106	[−0.015; 0.055]	1.021	0.113***	0.022	5.062	[0.069; 0.157]	1.000
Job role	H7	39	−0.242***	0.071	−3.381	[−0.381; −0.101]	1.003	−0.110	0.063	−1.741	[−0.235; 0.014]	1.005
Work level/ position	H8	26	0.381**	0.144	2.643	[0.098; 0.664]	1.001	0.240*	0.118	0.202	[0.008; 0.472]	1.021
Organizational												
Industry	H9	69	0.254***	0.016	24.042	[0.234; 0.275]	1.007	0.111*	0.048	2.287	[0.015; 0.205]	1.002
Control over AI	H10	66	−0.204***	0.007	−28.334	[−0.219; −.190]	1.003	−0.152***	0.035	−4.239	[−0.222; −0.081]	1.001
AI task and knowledge	H11	68	−0.010	0.008	−1.203	[−0.027; 0.006]	1.005	0.146**	0.047	3.071	[0.052; 0.239]	1.002
Macro-contextual												
Year	H12	71	−0.091***	0.007	−12.960	[−0.105; −0.077]	1.002	0.026	0.338	0.793	[−0.039; 0.093]	1.001
Country	H13	71	0.108***	0.013	8.373	[0.083; 0.134]	1.009	−0.016	0.059	−0.274	[−0.132; 0.099]	1.003
R^2			0.326					0.539				
AICc			108.490					40.279				

Abbreviations: β = beta; SE = standard error; CI = confidence interval; VIF = variance inflation factor; R^2 = coefficient of determination deviation; AICc = corrected Akaike Information Criteria
* $\rho < 0.1$; ** $\rho < 0.05$; *** $\rho < 0.001$

Table 3
Descriptive statistics.

	No. of effect sizes	Sample size	Simple mean r	Corrected mean r	SE	Z-value	$CI_{95\ \%}$	FSN	Egger's test	Q	I^2
Perceived Benefits of AI – AI adoption	168	19,215	0.266	0.324***	0.040	8.059	0.245; 0.403	2602	0.071	18,629.043***	99.00 %
Perceived Benefits of AI – Job engagement	137	14,475	0.273	0.315***	0.033	9.305	0.248; 0.381	2857	0.768	8227.815***	98.34 %

Abbreviations: r = correlation coefficient; SE = standard error; CI = confidence interval; FSN = Fail-safe Number; Q = weighted squared deviation, I^2 = the degree of heterogeneity
*** $\rho < 0.001$.

Egger's regression test values are insignificant, indicating minimal publication bias (Mehta et al., 2022). Fail-safe N values, significantly exceeding the number of effect sizes used in the meta-analysis, suggest minimal likelihood of bias (Rabiul et al., 2023). Specifically, 2602 and 2857 studies are required to nullify meta-analytic results on perceived benefits' relationship with AI adoption and job engagement, respectively.

Heterogeneity analysis results show significant Q values, confirming substantial variance in effect sizes. I2 values exceeding 80 % suggest a strong moderating effect of user-level, organizational and macro-contextual factors on perceived benefits' relationship with AI adoption and engagement (Maseeh et al., 2021).

4.2. Results of meta-regression

The meta-regression results further clarify the moderating influences on the relationships under investigation. Table 4 show that gender does not significantly moderate the relationship between perceived benefits and either AI adoption (β = 0.289, NS) or job engagement (β = 0.140, NS), rejecting H3a and H3b.

Educational attainment shows a negative moderating effect on AI adoption (β = −0.046*), supporting H4a, but not on job engagement (β = −0.034, NS), rejecting H4b.

Age positively moderates adoption (β = 0.012***), supporting H5a, but not engagement (β = −0.003), rejecting H5b.

Length of service does not influence adoption (β = 0.020, NS) but positively affects engagement (β = 0.113***), supporting H6b but not H6a.

Job role negatively moderates adoption (β = −0.242***), suggesting that more specialised or non-frontline roles perceive less benefit, supporting H7a. It does not significantly influence engagement (β = −0.110, NS), rejecting H7b.

Work position (e.g., managerial vs. non-managerial) positively moderates both adoption (β = 0.381**) and engagement (β = 0.240*), supporting H8a and H8b. Importantly, type of industry significantly and positively moderates both AI adoption (β = 0.254***) and engagement (β = 0.110*), supporting H9a and H9b. This suggests that service-centric industries like hospitality see stronger benefits from AI, likely due to the alignment of AI with customer service goals.

A notable unexpected finding is that control over AI negatively moderates both adoption (β = −0.204***) and engagement (β = −0.152***), supporting H10a and H10b, but contradicting earlier assumptions. This indicate that too much control may overwhelm or burden employees, reducing their perceived benefits, a point explored in more detail in the discussion section.

AI task and knowledge capabilities do not significantly moderate adoption (β = −0.010, NS), rejecting H11a, but positively influence engagement (β = 0.146**), supporting H11b.

Year of publication moderates adoption (β = −0.091***), supporting H12a, but not engagement (β = 0.026, NS), rejecting H12b.

Country context positively moderates adoption (β = 0.108***), supporting H13a, but not engagement (β = −0.016, NS), rejecting H13b.

Together, these moderators explain 32.6 % of the variance in perceived benefits' relationship with AI adoption and 53.9 % for job engagement, underlining their substantial explanatory power.

5. Discussions

5.1. Theoretical implications

This meta-analysis contributes to hospitality management research by addressing the relationship between frontline employees' perceived benefits of AI and their behavioural responses, specifically adoption and job engagement. While prior studies examined AI benefits on consumers (Gao et al., 2016; Shankar et al., 2024), employees (Gkinko and Elbanna, 2022), and organizations (Maragno et al., 2023), inconsistent findings and a lack of sector-specific meta-analyses have left key questions unresolved. This synthesis addresses that gap by integrating findings across multiple studies and identifying contextual and individual-level moderators relevant to hospitality work environments (Maseeh et al., 2021).

Findings align with SST, linking technology to the interplay of social, organizational, and technological factors rather than being purely technologically determined (Mackay and Gillespie, 1992; Russell and Williams, 2002). Results confirm that user-level, organizational and macro-contextual factors significantly affect how frontline employees perceive AI's value and how they respond to its integration, reinforcing SST's emphasis on user agency and social structure (Madan and Ashok, 2024).

At the user level, several individual characteristics moderated the relationship between perceived AI benefits and behavioural responses. Gender was not a significant moderator for either outcome (H3a, H3b), challenging assumptions in prior literature that gender differences influence technology acceptance and highlighting the potentially homogenising effects of hospitality work environments on employee perceptions (Li et al., 2024b). For instance, a front-desk agent's engagement with AI may relate more to perceived task efficiency than to personal background (Wijayati et al., 2022). Education negatively moderated adoption (H4a), suggesting that higher educational attainment does not necessarily facilitate AI uptake among frontline hospitality employees, potentially due to standardised task structures that minimise the differentiation of skills (Rabiul et al., 2023). However, education did not moderate engagement (H4b). Similarly, age positively moderated adoption (H5a) but not engagement (H5b), indicating that younger employees are more inclined to adopt AI, yet age differences do not significantly affect long-term engagement (Fu et al., 2024). Length of service did not moderate adoption (H6a) but had a positive effect on engagement (H6b), suggesting that experienced employees engage more deeply with AI over time, possibly due to their embeddedness in organizational routines (Frick et al., 2021; Shamim et al., 2023; Nguyen and Malik, 2022a). Additionally, job role negatively moderated adoption (H7a) but not engagement (H7b), indicating that frontline versus back-end distinctions affect the initial uptake of AI but not necessarily sustained interaction (Mirbabaie et al., 2022). Work level/position positively moderated both adoption (H8a) and engagement (H8b), suggesting that frontline employees in supervisory roles showed stronger alignment with AI benefits due to their need for real-time decision-making and direct service delivery (Park and Hyounae, 2020). These findings reinforce SST's premise that technology adoption is

Table 4
Summary of Hypothesis Outcomes.

Hypotheses (independent variable → outcome variable)		Corrected mean r	Significance level	Effect Directions
H1	*Perceived benefits of AI positively correlate with AI adoption.*	0.324	Highly significant	Positive
H2	*Perceived benefits of AI positively correlate with job engagement.*	0.315	Highly significant	Positive
Hypotheses (moderating variables)		**β**	Significance level	Effect Directions
H3a	*Gender moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in men.*	0.289	Not statistically significant	-
H3b	*Gender moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in men.*	0.140	Not statistically significant	-
H4a	*Educational attainment moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in bachelor's degree holders.*	−0.046	Marginally significant	Negative
H4b	*Educational attainment moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in bachelor's degree holders.*	−0.034	Not statistically significant	-
H5a	*Age moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in younger employees.*	0.012	Highly significant	Positive
H5b	*Age moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in younger employees.*	−0.003	Not statistically significant	-
H6a	*Length of service moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in tenured workers.*	0.020	Not statistically significant	-
H6b	*Length of service moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in tenured workers.*	0.113	Highly significant	Positive
H7a	*Job role moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in frontline roles.*	−0.242	Highly significant	Negative
H7b	*Job role moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in frontline roles.*	−0.110	Not statistically significant	-
H8a	*Work level moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in managerial roles.*	0.381	Statistically significant	Positive
H8b	*Work level moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in managerial roles.*	0.240	Marginally significant	Positive
H9a	*Type of industry moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in the service sector than in the non-service sector.*	0.254	Highly significant	Positive
H9b	*Type of industry moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in the service sector than in the non-service sector.*	0.110	Marginally significant	Positive
H10a	*Control moderates the relationship between AI benefits and adoption, with stronger effects in employees with more control.*	−0.204	Highly significant	Negative
H10b	*Control moderates the relationship between AI benefits and job engagement, with stronger effects in employees with more control.*	−0.152	Highly significant	Negative
H11a	*AI task and knowledge moderates the relationship between perceived benefits of AI and AI adoption, with a stronger effect in task-based capabilities.*	−0.010	Not statistically significant	-
H11b	*AI task and knowledge moderates the relationship between perceived benefits of AI and job engagement, with a stronger effect in task-based capabilities.*	0.146	**Statistically significant**	Positive
H12a	*Year moderates the relationship between AI benefits and adoption, with stronger effects in recent years.*	−0.091	**Highly significant**	Negative
H12b	*Year moderates the relationship between AI benefits and job engagement, with stronger effects in recent years.*	0.026	**Not statistically significant**	-
H13a	*Country moderates the relationship between AI benefits and adoption, with stronger effects in developed nations.*	0.108	**Highly significant**	Positive
H13b	*Country moderates the relationship between AI benefits and job engagement, with stronger effects in developed nations.*	−0.016	**Not statistically significant**	-

shaped by workplace context and role demands (Russell and Williams, 2002; Williams and Edge, 1996).

Industry type moderated both adoption (H5a) and engagement (H5b), confirming that hospitality's high-touch, customer-centric operations (e.g., concierge, front-desk agent, or restaurant server) uniquely position AI as both a functional and relational tool compared to efficiency-driven industries like retail (Chowdhury et al., 2022; Hu et al., 2021; Park and Hyounae, 2020). This industry-specific dynamic suggests that employees perceive and interact with AI differently depending on the sector's service logic, reinforcing the importance of sectoral context in SST applications (Russell and Williams, 2002; Williams and Edge, 1996). Control over AI emerged as a critical moderator, positively influencing both adoption (H10a) and engagement (H10b). This underscores SST's emphasis on agency where employees who can control and influence AI functionalities (e.g., being able to override chatbot recommendations or customise service workflows) exhibited greater adoption and engagement (Kang et al., 2023; Mirbabaie et al., 2022; Williams and Edge, 1996). AI task and knowledge significantly moderated job engagement (H11b) but not adoption (H11a). This suggests that employees are more likely to engage with AI when it enhances decision-making tasks (e.g. room assignments based on guest profiles), but adoption may lag without clear policies or managerial support (Howcroft and Taylor, 2023).

At the macro-contextual level, the publication year moderated the relationship between perceived benefits and adoption (H12a), reflecting increased AI receptivity during periods of accelerated digital transformation, such as the COVID-19 pandemic (Parvez et al., 2022; Wijayati et al., 2022). However, this temporal factor did not significantly affect engagement (H12b), reinforcing the notion that enduring factors like organizational culture and task design govern ongoing engagement (Park et al., 2024). Country context positively moderated adoption (H13a) but not engagement (H13b), indicating that institutional and infrastructural readiness in developed countries fosters AI uptake but does not uniformly translate to deeper employee engagement (Nguyen and Malik, 2022a).

The results extend the SST framework by demonstrating that individual characteristics, organizational factors, and socio-economic contexts shape the mechanisms through which perceived AI benefits influence adoption and engagement (Madan and Ashok, 2024). In the hospitality sector, where service delivery is closely tied to human-AI interaction, these findings advance theoretical understanding by emphasising that technology outcomes are contingent on employee

agency, task complexity, and structural conditions within organizations.

5.2. Managerial implications

The findings emphasise the importance of considering user-level, organizational and macro-contextual factors influencing AI adoption and job engagement, enabling managers to optimise AI benefits (Table 5). Timing and social context are critical for adoption (H12). For instance, during COVID-19, AI-supported contactless services (e.g., mobile check-ins and sanitation bots) gained traction (Leung et al., 2023; Parvez et al., 2022). Managers should monitor such trends and align strategies with evolving operational conditions.

Industry-specific and location-based strategies are essential (H9, H13). AI integration varies across sectors; healthcare prioritises predictive diagnostics and personalised treatments (Cao et al., 2021), while hospitality enhances customer experiences through chatbots, recommendations, and booking systems (Nguyen et al., 2023). For instance, SnatchBot Booking Travel Template enable intelligent customer guidance through the booking process, revolutionising guest interactions without human intervention (Abdelhakim et al., 2023; Barten, 2025). Similarly, AI-powered chatbots like ORAI's "Oshi" provide multilingual support, automate bookings, and personalise guest communications across multiple digital platforms, easing the workload of service staff and increasing guest satisfaction (Nguyen and Malik, 2022b; Peng et al., 2024; Barten, 2025). These tools gather valuable customer data to inform human agents and improve service continuity, highlighting the need for managers to align AI approaches with sector and regional demands to maximise effectiveness (Loureiro et al., 2023; Verma and Singh, 2023).

Country-specific factors (H13), including socio-economic conditions, institutional frameworks, and policies, should also guide AI integration to overcome structural barriers in developing regions and optimise scalability in developed markets (Wu et al., 2024).

Empowering frontline employees with appropriate levels of AI control improves behavioural outcomes (H10). Practical measures such as involving frontline employees in AI implementation pilots, offering skill-based training, and creating structured feedback loops help build ownership, reduce resistance, and facilitate smoother integration (H7, H10) (Lin et al., 2022). However, analysis showed that too much control over AI can overwhelm employees rather than empower them, particularly in fast-paced hospitality settings, thereby reducing both adoption and engagement (H10). Job roles benefit when AI assists rather than burdens tasks (H7). For example, AI-driven scheduling platforms such as 7shifts, Rippling, and Shiftboard (Behnke, 2025), automate staff rostering and task coordination, enabling teams to focus on personalised guest experiences (Wong et al., 2023).

Work level/position significantly moderated adoption and engagement (H8), with managers and supervisors showing higher receptivity to AI's potential (Cao et al., 2021). Equipping leaders with dashboard-based analytics and service optimisation tools fosters data-driven leadership that can champion AI adoption across operational teams (Frick et al., 2021). Educational attainment negatively influenced adoption (H4), suggesting that highly educated employees may perceive operational AI as diminishing the value of expertise (Kong et al., 2021; Suseno et al., 2023). Emphasising AI's role (H11) in enhancing strategic decision-making and guest customisation can mitigate this, particularly in roles like concierge services (Yu and Li, 2022). Age positively influenced adoption (H5), indicating that older employees can adopt AI effectively when supported with relevant training and intuitive tools (Li et al., 2024b). Length of service enhanced engagement (H6), reinforcing the importance of experienced staff in sustaining AI use when adequately supported (Frick et al., 2021; Presbitero and Teng-Calleja, 2023).

Gender and job role showed no significant moderation on engagement (H3, H7), indicating limited value in demographic targeting. Instead, structuring interventions around organisational roles and support systems is more effective (Mirbabaie et al., 2022). In hospitality businesses like Solotel, website chatbots manage routine enquiries while AI-driven pricing tools and promotional strategies have enhanced guest experiences and increased revenue (Dequito, 2024). These tailored AI applications boost customer engagement and allowed staff to focus on complex or high-touch services (Michel et al., 2023; Qiu et al., 2022).

Table 5
Practical Checklist for AI Readiness in Hospitality.

Domain	Managerial Consideration
Strategic Fit	Align AI with sector goals by enhancing guest experiences through chatbots, booking systems, service automation, and tailored promotional strategies (Parvez et al., 2022; Peng et al., 2024; Dequito, 2024; Loureiro et al., 2023; Verma and Singh, 2023)
Role Clarity	Match AI tools to job roles (e.g., use task automation for service staff, AI scheduling for operations, and analytics dashboards for supervisors) (Peng et al., 2024; Rabiul et al., 2023; Zirar et al., 2023).
Employee Readiness	Tailor training to employee profiles. Support older staff with intuitive tools and frame AI as a strategic asset to engage highly educated employees (Ding, 2021; Kim et al., 2021; Li et al., 2024b; Loureiro et al., 2023; Wang et al., 2023b).
Technical Infrastructure	Integrate AI systems with existing customer relationship management to streamline data flow, improve service continuity, and coordinate human-AI interactions (Parvez et al., 2022; Wu et al., 2024).
Feedback Mechanisms	Implement structured feedback via surveys and participatory sessions to foster ownership and refine AI deployment (Mackay and Gillespie, 1992; Madan and Ashok, 2024).
Trust & Ethics	Involve frontline staff in pilots and decision-making to build trust and reduce automation-related concerns (Loureiro et al., 2023; Madan and Ashok, 2024).
Scalability	Design AI strategies that consider socio-economic conditions and institutional frameworks for scalable, sustainable integration across regions (Wu et al., 2024; Zirar et al., 2023).

Effective AI strategies in hospitality require role-appropriate deployment, clear communication of AI's value, and balanced autonomy (Mirbabaie et al., 2022; Tojib et al., 2023). Tailored training, alignment of AI applications with job-specific needs, and ongoing feedback mechanisms (e.g., providing dashboard-based analytics to supervisors and task automation tools to waitstaff) will maximise adoption and engagement, ensuring AI integration enriches the hospitality service experience (Nguyen and Malik, 2022a).

5.3. Limitations

This meta-analysis offers valuable insights but has limitations. First, the literature search was restricted to English-language articles, introducing potential language bias and limiting generalisability across non-English-speaking regions (Jebarajakirthy et al., 2024). Second, it highlights the significance of publication year as a moderator, especially during COVID-19 (e.g., (Kim et al., 2021); Leung et al., 2023; Wu et al., 2024). However, rapid changes in AI adoption and job engagement in hospitality may render findings outdated (Adil et al., 2022). Third, although this study briefly addresses ethical and privacy concerns related to AI, deeper exploration is warranted in hospitality sectors where issues such as surveillance, data consent, and algorithmic bias may affect both frontline employee trust and customer experience (Malik et al., 2021; Rodgers et al., 2023; Wang et al., 2023b). For instance, Disneyland's use of facial recognition to streamline guest entry raises privacy and consent concerns (Detres, 2025), highlighting the ethical complexities of integrating AI in customer-facing environments. Finally, a lack of longitudinal evidence in the included studies restricts understanding of long-term behavioural changes (Abdelhakim et al., 2023). Future research should adopt longitudinal designs to capture sustained impacts of AI over time (Gkinko and Elbanna, 2022).

5.4. Future research directions

This meta-analysis highlights relationships between frontline employees' perceived benefits of AI and outcome variables, identifying key areas for advancing research on AI's perceived benefits in service delivery. Future research directions are outlined by constructs, moderation, methodology and interdisciplinary approaches (Table 6):

5.4.1. Constructs

5.4.1.1. Expanding constructs. Future studies should incorporate constructs like organizational culture (Lingmont and Alexiou, 2020), resilience (Teng et al., 2024), innovation capability (Peng et al., 2024), and knowledge sharing (Lim and Ok, 2021) to gain a holistic view of AI's behavioural impact.

5.4.1.2. Address negative constructs. In addition to positive outcomes, future studies should examine negative constructs such as burnout (Kong et al., 2021; Liang et al., 2022), job displacement (Tong et al., 2021), and ethical concerns (Wang et al., 2023b) to provide a balanced perspective and help mitigate AI-related risks.

5.4.1.3. Deconstructing adoption. Investigating AI adoption as a multidimensional process (e.g., implementation, usage, optimisation) may yield more targeted strategies (Li et al., 2024a).

5.4.1.4. Ethical and privacy considerations. In frontline service environments, future research should explore ethical and privacy implications of AI adoption more comprehensively (Malik et al., 2021; Wang et al., 2023b). Topics such as data transparency, employee surveillance, algorithmic fairness, and consent processes merit attention, influencing employee trust, psychological safety, and organizational legitimacy (Lim and Ok, 2021; Suseno et al., 2023; Yu and Li, 2022).

5.4.2. Moderation

5.4.2.1. Expand contextual variables. Future research could examine variables such as organizational readiness (Frick et al., 2021), digital literacy (Shamim et al., 2023), and team dynamics (Lin et al., 2024) to deepen understanding of AI integration.

5.4.2.2. Cross-cultural moderation. Examining cultural factors moderating AI adoption effects across regions (Leung et al., 2023) could reveal how technology attitudes vary globally, informing culturally sensitive interventions, such as ethical governance or emotional intelligence training (Khoa et al., 2023; Rodgers et al., 2023).

5.4.2.3. Include temporal moderation. As AI and robotics mature in hospitality (Khoa et al., 2023), time-based factors could moderate behavioural outcomes. Longitudinal studies can help trace evolving relationships between AI use and employee engagement (Prentice et al., 2023).

5.4.3. Methodology

5.4.3.1. Longitudinal studies. Longitudinal methods (Nguyen and Malik, 2022b) can capture the dynamic nature of AI adoption and its impacts (Castaneda et al., 2025; Lin et al., 2024), revealing long-term influences

Table 6
Future Research Priorities.

Priority Level	Research Focus	Description	Reason for Priority
High Priority *(Immediate)*	Longitudinal Studies	Track changes in AI adoption and engagement over time to assess long-term impact (Castaneda et al., 2025; Lin et al., 2024; Nguyen and Malik, 2022b).	Urgent due to lack of long-term data and rapid tech shifts in AI implementation (Liang et al., 2022; Prentice et al., 2023).
High Priority *(Immediate)*	Ethical and Privacy Considerations	Study AI-related issues like surveillance, fairness, and consent in customer-facing industries (Lim and Ok, 2021; Suseno et al., 2023; Yu and Li, 2022).	Vital for trust-building and responsible AI adoption in high-contact settings like hospitality (Malik et al., 2021; Wang et al., 2023b).
High Priority *(Immediate)*	Cross-Cultural Moderation	Examine how regional and cultural differences shape AI attitudes and outcomes (Leung et al., 2023).	Crucial for global AI strategy alignment and culturally sensitive implementation (Khoa et al., 2023; Rodgers et al., 2023).
High Priority *(Immediate)*	Address Negative Constructs	Explore job burnout, displacement, and risk perception to balance optimistic outcomes (Kong et al., 2021; Liang et al., 2022; Tong et al., 2021).	Balances optimism with caution and supports development of mitigation strategies (Wang et al., 2023b).
Medium Priority *(Short-Term Feasible)*	Expand Contextual Variables	Study moderators like team dynamics, readiness, and digital literacy (Frick et al., 2021; Lin et al., 2024).	Enhances applied understanding of what drives AI adoption in different work environments (Shamim et al., 2023).
Medium Priority *(Short-Term Feasible)*	Expanding Constructs	Add variables such as innovation capability, knowledge sharing, and resilience (Lingmont and Alexiou, 2020; Lim and Ok, 2021; Teng et al., 2024).	Broadens the explanatory power of models by reflecting dynamic workplace behaviours (Peng et al., 2024).
Medium *(Conceptually Complex)*	Deconstructing AI Adoption	Analyse AI adoption as a process (i.e., implementation, use, and optimisation) with different influencing factors (Li et al., 2024a).	Offers targeted strategies across AI lifecycle stages rather than treating adoption as a single event (Li et al., 2024a).
Medium Priority *(Conceptually Complex)*	Include Temporal Moderation	Explore how timing and AI maturity affect behavioural responses over time (Khoa et al., 2023).	Adds nuance by capturing shifting perceptions and technology familiarity (Prentice et al., 2023).
Medium Priority *(Methodologically Intensive)*	Experimental Designs	Test specific interventions like training formats or varying AI control levels (Wijayati et al., 2022).	Enables causal inferences and strategic validation of implementation techniques (Wang et al., 2023b).
Low Priority *(Testing Phase)*	AI Simulation and Pilot Trials	Use small-scale trials to evaluate behavioural impacts before full-scale rollout (Lin et al., 2022; Yu and Li, 2022).	Useful for early-stage feedback and risk identification, especially in high-stakes service roles (Madan and Ashok, 2024).
Low Priority *(Long-Term Generalisability)*	Broader Geographic and Industry Samples	Include more diverse regions and sectors to improve representativeness (Chowdhury et al., 2022; Kambur and Akar, 2022).	Helps generalise AI findings across industries beyond hospitality and IT (Zhu and Kanjanamekanant, 2023).
Low Priority *(Method Enrichment)*	Mixed-Methods Approaches	Combine surveys and interviews to triangulate employee insights (Cichocki and Kuleshov, 2021).	Supports comprehensive exploration of subjective and quantitative outcomes (Cichocki and Kuleshov, 2021).
Low Priority *(Interdisciplinary Expansion)*	Interdisciplinary Approaches	Integrate AI ethics, behavioural science, and human-computer interaction frameworks (Verma and Singh, 2023; Wang et al., 2023b; Yu and Li, 2022).	Encourages responsible, human-centered design and enhances trust and user experience (Khoa et al., 2023).

on engagement and performance (Liang et al., 2022; Prentice et al., 2023).

5.4.3.2. Experimental designs. Experimental designs could test interventions affecting AI adoption and engagement (Wijayati et al., 2022), such as varying employee control levels over AI or training types (Wang et al., 2022).

5.4.3.3. Mixed-methods approaches. Combining methods, such as quantitative surveys with qualitative interviews, could provide richer insights (Cichocki and Kuleshov, 2021).

5.4.3.4. Broader geographic and industry samples. Expanding research to include broader geographic (Kambur and Akar, 2022) and industry samples (Chowdhury et al., 2022) can improve generalisability. For instance, while technology stress varies by industry, IT sectors may find it beneficial for well-being (Zhu and Kanjanamekanant, 2023). Researching AI adoption across untapped regions and industries could reveal context-specific challenges and opportunities.

5.4.3.5. AI simulation and pilot trials. Future studies could implement AI simulation environments or small-scale pilot trials to test frontline employee responses before full-scale deployment (Lin et al., 2022; Yu and Li, 2022). These methods can help identify implementation challenges, acceptance barriers, and behavioural impacts in controlled yet realistic settings (Madan and Ashok, 2024).

5.4.4. Interdisciplinary approaches

To deepen understanding of AI in frontline service contexts, future research should adopt interdisciplinary frameworks that integrate insights from multiple fields. First, AI ethics can provide valuable guidance on applying responsible AI principles in hospitality, shaping employees' behavioural intentions and attitudes toward AI use (Wang et al., 2023b). Second, behavioural science can help investigate how factors such as AI transparency influence human trust in AI systems, particularly in high-contact service environments (Yu and Li, 2022). Third, human-computer interaction offers theoretical foundations such as the "computers as social actors" paradigm, which suggests that people often respond to AI and digital systems in ways similar to interactions with other humans (Verma and Singh, 2023), generating practical strategies to reduce resistance, enhance trust, and promote ethical, user-centred behavioural outcomes (Khoa et al., 2023).

6. Conclusion

This meta-analysis synthesises existing perceived benefits of AI in frontline employees and introduces a framework showing how these perceptions influence adoption and engagement. The findings highlight that frontline engagement with AI is not uniform; it is moderated by user-level, organizational, and macro-contextual factors. Specifically, industry type, control over AI, and employee work level notably influence how AI is perceived and adopted (Maseeh et al., 2021).

The study reinforces the SST framework, illustrating that AI integration is not merely technical but shaped by social structures, workforce attributes, and contextual norms (Howcroft and Taylor, 2023). While the analysis includes multiple service industries, its emphasis on hospitality offers actionable insights into how AI can be aligned with employee competencies and operational goals (Hu et al., 2021).

For hospitality management, AI adoption strategies must be tailored to the roles, expectations, and values of frontline employees (Chandra and Rahman, 2024; Pillai et al., 2024; Song et al., 2022b). Building trust, offering control, and addressing ethical concerns are essential to sustaining employee engagement (Alam et al., 2024; Dutta et al., 2022; Shamim et al., 2023). This includes practical measures such as involving staff in AI implementation decisions (Yu and Li, 2022), using automation to reduce repetitive tasks like check-ins or inventory updates (Behnke, 2025; Dequito, 2024; Schwarz, 2025), and providing support for employees adapting to AI (Verma and Singh, 2023). Moreover, training programs should reflect varying comfort levels with technology, especially across age and experience levels, to avoid disengagement or resistance in service-driven environments (Frick et al., 2021; Tojib et al., 2023). With the hospitality AI market projected to grow from $90 million in 2023 to $8 billion by 2033, the industry is poised for significant transformation (Schwarz, 2025). Continued research is essential to track how evolving AI applications affect employee experiences, service quality, and organizational performance.

CRediT authorship contribution statement

Anne Rachelle Castaneda: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Haroon Iqbal Maseeh:** Validation, Supervision, Methodology, Data curation, Conceptualization. **Jiraporn Surachartkumtonkun:** Writing – review & editing, Supervision, Project administration. **Wei Shao:** Writing – review & editing, Supervision. **Park Thaichon:** Writing – review & editing, Supervision.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijhm.2025.104446.

Data availability

The research data and analysis code are included in the Supplementary Materials and considered part of the article's Appendices.

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